Module 3: Probability and Distributions

Five lessons | From event rules to probability models

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Learning sequence

Module 2 described recorded data. This module defines uncertainty about a selection or future outcome, then calculates probabilities under explicit models. Finish the descriptive-statistics review first.

1. **Probability rules:** outcomes, events, complements, intersections, and unions.
2. **Conditional probability:** reference groups, independence, multiplication, and Bayes' theorem.
3. **Discrete random variables:** a probability mass function, cumulative probabilities, expected value, and variance.
4. **Binomial and Poisson models:** assumptions, count probabilities, exposure, and simulation.
5. **Continuous distributions:** uniform areas, normal tails, standardization, and quantiles.

Lessons and practice

Lesson	Main question	Practice
1. Probability rules	What event and selection mechanism define the probability?	Lesson 1 practice
2. Conditional probability	How does new information change the reference group?	Lesson 2 practice
3. Discrete random variables	What values can occur, with what probabilities?	Lesson 3 practice
4. Binomial and Poisson models	Which count mechanism and event apply?	Lesson 4 practice
5. Continuous distributions	Which area or cutoff answers the question?	Lesson 5 practice

Finish with the integrated probability and distributions review. Separate instructor guides cover each practice and the review.

Cases, inputs, and readings

Use this guide to connect each case to its input type and selected textbook reading.

Case	Input used	Main 8th-edition reference
Restaurant selection	Existing checked 300-row CSV	Tables 2.9–2.10, pp.65–66; probability rules in §§4.1–4.4
Two suppliers	Stated shares and good/bad probabilities	§4.5, pp.201–205
DiCarlo Motors	Six published daily-sales frequency cells	§§5.2–5.3, pp.222–228
Martin Clothing Store	Three customers, purchase probability 0.30	§5.5, especially p.244
Arrivals	Mean 10 per 15 minutes and stated process assumptions	§5.6, pp.252–255
Flight duration	Uniform bounds 120–140 minutes	§6.1, pp.275–277
Grear tires	Normal mean 36,500 miles and SD 5,000 miles	§6.2, especially pp.286–290

The Restaurant CSV retains its documented provenance and verification limits. All other inputs are explicitly identified printed aggregate values or model parameters entered in the lesson code. They are not new individual-observation datasets. Artificial simulation is labeled as such.

Read each case's English introduction followed by its equivalent Arabic narration. Statistical explanations and interpretation tasks remain in English.

Scope and next step

The selected sequence includes five lessons, five practices, and an integrated review. It is not exhaustive coverage of Chapters 4–6: detailed permutation exercises, bivariate portfolio distributions, hypergeometric calculations, and the exponential distribution remain possible extensions.

Module 4: Sampling and estimation is the next topic; its four lessons, practice, and review are available. We will distinguish the distribution of an individual observation from the sampling distribution of a statistic.

Return to the course home.